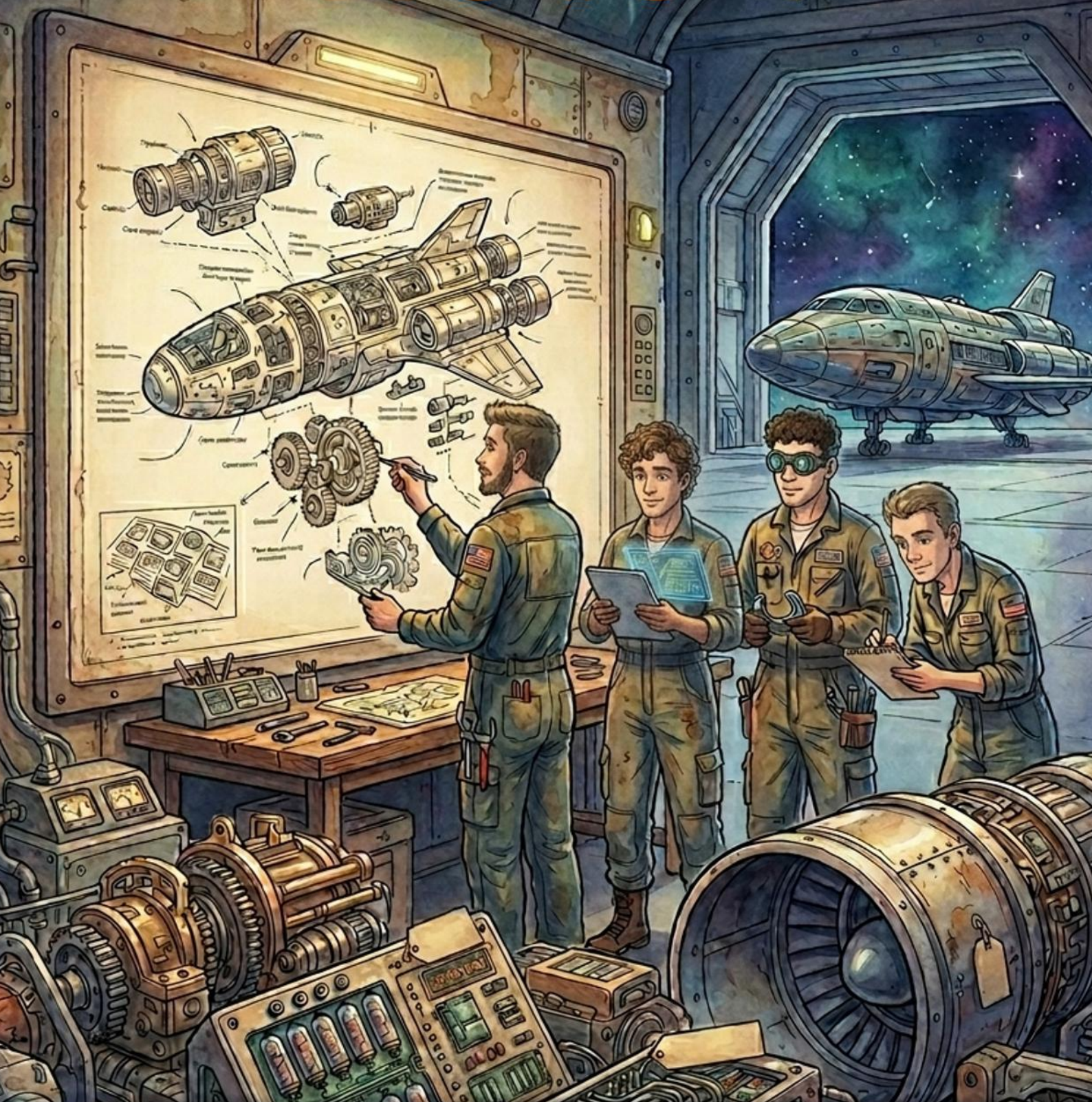


noted



iscte

UNIVERSITY
INSTITUTE
OF LISBON

Engineering of Intelligent Models

FINAL DELIVERY

SETUP AND INSTALLATION MANUAL

NOTED PLATFORM

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1. Introduction

This document is a short, focused walkthrough for getting **noted** running on your machine, with both demo projects (jena_weather and jena_client) configured as mounts, so you can exercise the full platform end-to-end.

If you only want to poke around without installing anything, a live instance is available at:

Live demo: <https://logus2k.com/noted>

For a permanent setup, follow the steps below.

Quick links

- **noted source:** <https://github.com/logus2k/noted>
- **jena_weather:** https://github.com/logus2k/jena_weather (training project): clone from the tutorial materials you received.
- **jena_client:** https://github.com/logus2k/jena_client (serving client demo): clone from the tutorial materials you received.
- **User Manual:** documents/user-manual/ in the noted repo, or inside noted itself under **Knowledge Base > noted User Manual** once the platform is running.

2. Prerequisites

- **Docker** (Engine 24+ or Desktop 4.30+).
- **Docker Compose** plugin (docker compose subcommand).
- **NVIDIA Container Toolkit** (optional, required only if you want GPU acceleration for training). See:
<https://docs.nvidia.com/datacenter/cloud-native/container-toolkit/latest/install-guide.html>
- **Git** for cloning the repositories.
- **~20 GB free disk** for Docker images and model artifacts.
- A **Linux host** is recommended. macOS and Windows work via Docker Desktop but the GPU path is Linux-only.

3. Clone the repositories

Clone all three repositories to a common parent directory, for example ~/noted-demo/:

```
mkdir -p ~/noted-demo && cd ~/noted-demo

# noted platform
git clone https://github.com/logus2k/noted.git

# Demo training project
git clone <your-jena_weather-repo-url> jena_weather

# Demo serving client
git clone <your-jena_client-repo-url> jena_client
```

Your layout should now look like:

```
~/noted-demo/
├─ noted/
├─ jena_weather/
└─ jena_client/
```

4. Configure environment variables

noted reads Docker Compose environment variables from `services/.env`. A template is provided.

```
cd ~/noted-demo/noted/services
cp .env.example .env
```

Open `.env` and set the following:

```
# Required: a shared secret the noted terminal uses to authenticate
# terminal sessions. Any non-empty string works for local review.
NOTED_TERMINAL_SECRET=changeme

# Optional: enables Claude models in the noted Assistant.
# Without this, only the local Gemma 4 model is available.
# Get a key at https://console.anthropic.com/
ANTHROPIC_API_KEY=

# Airflow credentials (default admin login for the Airflow UI)
_AIRFLOW_WWW_USER_USERNAME=airflow
_AIRFLOW_WWW_USER_PASSWORD=airflow

# MinIO credentials (the artifact store backing MLflow)
MINIO_ROOT_USER=admin
MINIO_ROOT_PASSWORD=password
```

The other fields in the template can be left at their defaults for a local review.

5. Configure project mounts

Important — read this whole section: the two demo projects (jena_weather and jena_client) must be configured as **mounts** in noted before you can test the platform end-to-end. Otherwise noted starts up empty and none of the tutorial flows work.

noted's mounts are configured in the file data/NOTED.md, which uses YAML frontmatter to declare host paths that will be bind-mounted into the noted container. Edit that file before the first launch:

```
cd ~/noted-demo/noted
nano data/NOTED.md  # or your editor of choice
```

Replace the frontmatter section at the top so it matches the absolute host paths of the two cloned demo projects. For the example layout above:

```
---
mounts:
- name: jena_weather
  host_path: /home/youruser/noted-demo/jena_weather
- name: jena_client
  host_path: /home/youruser/noted-demo/jena_client
---

# Noted Configuration

Edit this file to manage your workspace mounts.
(rest of the file can stay as-is)
```

Replace /home/youruser/noted-demo/... with your real absolute paths. Relative paths are **not** supported - the file is read by Docker Compose, which needs absolute host paths.

When noted starts up, it auto-generates data/docker-compose.mounts.yml from NOTED.md. You do not need to edit that file by hand - it is regenerated on every start.

If you forget this step before the first launch, you can still add mounts later via the **Add Mount** action in the Explorer UI: open the Explorer sidebar, click **Mounts**, and use the context menu. The UI writes the new entry into data/NOTED.md for you. But for the demo projects specifically, configuring them before the first launch is faster and cleaner.

6. Launch noted and its dependencies

With GPU

```
cd ~/noted-demo/noted/services

docker compose \
  -f docker-compose.yml \
  -f docker-compose.gpu.yml \
  -f ../data/docker-compose.mounts.yml \
  up -d --build
```

CPU only

```
cd ~/noted-demo/noted/services

docker compose \
  -f docker-compose.yml \
  -f ../data/docker-compose.mounts.yml \
  up -d --build
```

The first launch builds several images and may take 5-15 minutes depending on your network and CPU. Subsequent launches reuse cached layers.

On success, docker compose ps shows the containers running:

```
noted                Up (healthy)
noted-serving        Up (healthy)
noted-graph          Up (healthy)
noted-evidently      Up (healthy)
noted-nginx          Up
mlflow               Up (healthy)
minio                Up (healthy)
postgres             Up (healthy)
redis                Up (healthy)
airflow-apiserver    Up (healthy)
airflow-scheduler    Up (healthy)
airflow-worker       Up (healthy)
... (more airflow workers)
```


7. Open noted

Open your browser at:

<http://localhost:8123>

You should see the noted interface with:

- An Explorer sidebar on the left with root sections (Projects, Experiments, Data, Orchestration, Models, Environments, Assistant, Knowledge Base).
- Under **Mounts** (inside the Projects tree), your jena_weather and jena_client projects, ready to explore.
- A **Knowledge Base** > **noted User Manual** category with 7 pages covering every aspect of the platform.

8. First-run checklist

A quick smoke test to confirm everything is wired up correctly:

1. Open **Knowledge Base > noted User Manual > 1. Your First Project** and follow Page 1. This verifies project creation, environment creation, notebook, and kernel are working.
2. Open **Mounts > jena_weather > notebooks > emi_tutorial3_jena_weather.ipynb** and run a single cell using the Run Manager. This verifies Hydra config injection, MLflow tracking, and DVC data lineage end-to-end.
3. Click **Experiments** in the icon bar, drill down to the run you just created, and verify that all 5 lineage badges (Data, Config, Code, Run, Model) are populated.
4. Register the run's model, deploy it via the **Deploy** button, and click **Try It** to send a synthetic prediction. This verifies the serving stack.
5. Launch the `jena_client` standalone web app from a noted terminal (see the `jena_client` README inside its repo) and load a model via its dropdown UI. This verifies the external client path.

If all 5 steps pass, your review environment is fully functional.

9. Stopping and cleaning up

Stop (preserves data and images)

```
cd ~/noted-demo/noted/services
docker compose \
  -f docker-compose.yml \
  -f docker-compose.gpu.yml \
  -f ../data/docker-compose.mounts.yml \
  down
```

Stop and wipe volumes (removes MLflow runs, MinIO objects, Airflow DB)

```
docker compose \
  -f docker-compose.yml \
  -f docker-compose.gpu.yml \
  -f ../data/docker-compose.mounts.yml \
  down -v
```

Remove the noted image (frees disk)

```
docker rmi logus2k/noted
```

The data/ directory on your host is **not** touched by any down command. It contains your projects, notebooks, environments, and the noted configuration. Delete it manually only when you want a completely fresh start.

10. Troubleshooting

noted is unreachable at <http://localhost:8123>

```
docker compose ps           # is the noted container running?
docker logs noted | tail -60 # what does it say?
```

Most commonly a port conflict: another process is already on 8123. Stop it or change the exposed port in docker-compose.yml.

“permission denied” on data/NOTED.md when editing

The file may be owned by root from a previous container run. `sudo chown $USER:$USER data/NOTED.md` or edit with `sudo`.

Mounts do not appear in the Explorer

Check that data/NOTED.md has absolute host paths and that the paths actually exist on the host. Run `docker exec noted ls /app/mounts/` to confirm noted sees them. If the mounts file was edited after the container started, restart the noted service:

```
docker compose \
  -f docker-compose.yml \
  -f docker-compose.gpu.yml \
  -f ../data/docker-compose.mounts.yml \
  up -d --force-recreate --no-deps noted
```

GPU not detected inside the noted container

Verify the NVIDIA Container Toolkit is installed and working on the host: `docker run --rm --gpus all nvidia/cuda:12.4.0-base-ubuntu22.04 nvidia-smi` should list your GPU. If that succeeds but noted still does not see the GPU, confirm you used the `-f docker-compose.gpu.yml` override on the docker compose up command.

Assistant dropdown shows only the local model

Set `ANTHROPIC_API_KEY` in `services/.env` and restart the noted container with `--force-recreate --no-deps noted`.

11. Next steps

- Follow the 7-page **User Manual** (Knowledge Base > noted User Manual) for a guided tour of every capability.
- Read the project documentation in documents/ for architectural and planning context: noted_plan.md, noted_vision.md, noted_scope.md, and README.md.
- Ask questions inside the platform using the **noted Assistant** panel - it has 40+ skills covering Airflow, DVC, Evidently, Hydra, MLflow, and noted itself.